llmXive Automated Review of Crafter: A Multi-Agent Harness for Editable Scientific Figure Generation from Diverse Inputs

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their support by the provided evidence.

1. Ambiguity in Baseline Comparisons (Abstract, Section 1, Section 4.1) The abstract and Section 1 claim that *Crafter* "substantially outperforms... the agentic baseline." Section 4.1 clarifies this refers to a "controlled comparison" where all agentic methods use the same backbone (Nano Banana 2), yielding a 16.61 point gap on PaperBananaBench. However, the abstract omits the "controlled comparison" qualifier. Without it, a reader might compare *Crafter* (w/ Nano Banana 2) against the strongest available baseline, *PaperBanana* (w/ Nano Banana Pro), where the gap is smaller (14.38 points). While *Crafter* still wins, the claim of "substantial" outperformance is slightly inflated if the backbone difference is ignored. The abstract should explicitly state "under controlled comparison" to ensure the claim accurately reflects the experimental design.

2. Overstatement of Human-VLM Agreement (Section 3.1, Appendix A.8) Section 3.1 states: "A blind human study... confirms that this metric tracks human preference." Appendix A.8 reports a Cohen's kappa of 0.58. In standard statistical interpretation, 0.58 represents "moderate" agreement, not "substantial" or "strong." While 72% raw agreement is positive, the claim implies a stronger validation than the data supports. The text should be revised to reflect the "moderate" nature of the agreement to avoid overstating the reliability of the VLM judge.

3. Precision of "No Baseline Surpasses" Claim (Section 4.1) The text states, "No baseline surpasses Crafter in any column." This is technically true for the specific rows presented in Table 1. However, the table includes baselines with different backbones (e.g., PaperBanana w/ Nano Banana Pro). The claim could be misinterpreted as a universal dominance across all configurations. The text should clarify that this holds specifically within the "controlled comparison" setting where backbones are matched, ensuring the claim is not read as a general statement about all possible baseline configurations.

Conclusion The core scientific claims are supported by the data, but the phrasing in the abstract and results section requires minor precision adjustments to accurately reflect the experimental constraints (backbone matching) and the statistical strength of the human evaluation (moderate kappa).

Action items.

- [writing] Abstract claims 'substantially outperforms... agentic baseline' without specifying 'controlled comparison' (same backbone). Table 1 shows the gap shrinks if comparing against PaperBanana (w/ Nano Banana Pro). Clarify the baseline definition in the abstract to avoid ambiguity.
- [writing] Section 3.1 claims the VLM metric 'tracks human preference' based on a study with Cohen's kappa=0.58. This indicates 'moderate' agreement, not strong validation. Temper the claim to reflect the moderate agreement level accurately.
- [writing] Section 4.1 states 'No baseline surpasses Crafter in any column'. While true for the specific rows shown, the comparison mixes backbones (Nano Banana 2 vs Pro) in

the full table. Ensure the text explicitly limits this claim to the 'controlled comparison' setting to prevent misinterpretation.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear and engaging visual overview of the multi-agent architecture. However, there are minor inconsistencies between the specific variable names in the caption (e.g., S_0) and the labels in the diagram (S), and the input terminology differs slightly.

Figure 2

The figure provides a clear and readable visual workflow for the three-phase pipeline. However, the caption is marred by unresolved LaTeX formatting artifacts and incomplete sentences that hinder comprehension.

Figure 3

The figure visually presents four distinct task types but the caption is incomplete, containing a LaTeX formatting error and failing to define the specific tasks shown in the columns.

Figure 4

The figure presents a clear qualitative comparison grid, but the caption is insufficient as it fails to define the column headers (methods) and row labels (input conditions) visible in the image.

Figure 5

The figure fails to support the caption's claim of showing three task constructions, displaying only a single raw UI screenshot with extraneous interface elements that obscure the scientific content.

Figure 6

Figure 6 accurately depicts the blind pairwise human-evaluation interface described in its caption, showing the conditioning input at the top and the randomized Figure A/B comparison below. The layout is clear, legible, and effectively demonstrates the data collection process for the study.

Figure 7

The figure effectively displays a qualitative comparison of four systems, but the caption is defective with missing column labels and lacks a definition for the numerical scores shown.

Figure 8

The figure is fundamentally broken as the visual content of the 'Ours' column contradicts the caption's description of 'failure cases' (showing perfect diagrams instead of errors) and the input text does not match the architectural diagrams shown in the output columns.
Action items.

- [writing] Figure 1: The caption states 'Given context and Docs', but the diagram explicitly labels the input as 'Input Context' (speech bubble) and 'Document' (paper icon), creating a minor terminology mismatch.
- [writing] Figure 1: The caption mentions 'seeds S_0', but the diagram labels the central scroll simply as 'S' without the subscript '0', which may cause confusion regarding the iteration state.
- [writing] Figure 2: The caption contains LaTeX placeholders (e.g., 'a^*', 'v', '()') and incomplete sentences (e.g., 'verifies the cleaned canvas ()') that are not resolved by the visual content, making the text difficult to read.
- [writing] Figure 2: The caption describes a 'VLM designer D' and 'image editor E', but the figure labels these roles as 'Vision-Language Designer Agent' and 'Image-Editing Executor' without explicitly mapping them to the variables D and E used in the text.
- [science] Figure 3: The caption states 'Each column shows one task,' but the image displays four distinct columns labeled (a) Text-to-image, (b) Mask-completion, (c) Key-element, and (d) Sketch. The caption fails to define or describe these specific task categories, making the figure's content ambiguous without external context.
- [writing] Figure 3: The placeholder text 'Representative \ samples' in the caption contains a LaTeX formatting error (missing command name), which should be corrected to 'Representative Crafter samples' or similar for clarity.
- [writing] Figure 4: The column headers ('Conditioning input', 'PaperBanana', 'AutoFigure', 'Ours', 'Ground truth') are not defined in the caption, which only states 'Qualitative comparison across different input conditions'.

- [writing] Figure 4: The row labels ('Text-to-image', 'Mask completion', 'Key-element composition', 'Sketch conditioned') are not defined in the caption, making the specific input conditions ambiguous.
- [science] Figure 5: The caption claims to show 'three reference-conditioned task constructions' and 'three graduate-level annotators,' but the image displays only a single interface instance with no evidence of the three distinct task types or the multi-annotator workflow described.
- [writing] Figure 5: The image is a raw screenshot of an annotation tool UI (including 'Close region picker', sliders, and 'Apply edits to disk' buttons) rather than a polished figure illustrating the 'task constructions' or 'drawing tools' mentioned in the caption.
- [writing] Figure 7: The caption contains missing text for the column headers ('Columns: input raster, , , \ (green frame)'), failing to name the 'Edit-Banana' and 'AutoFigure-Edit' systems shown in the image.
- [writing] Figure 7: The caption states 'Per-panel numbers are three-VLM judge means' but does not define the scale or units (e.g., 0-10) for these scores.
- [science] Figure 8: The 'Ours' column (red frame) displays a fully rendered, high-quality diagram with detailed text and arrows, which directly contradicts the caption's claim that this figure represents 'failure cases' (specifically 'dropped panels', 'mismatched infill', and 'literal skeleton'). The visual content of the 'Ours' column does not match the failure modes described in the caption or the labels on the left.
- [science] Figure 8: The 'Conditioning input' column contains a text block describing a 'multi-dimensional task structure' and 'experimental procedure', but the corresponding 'Ours' and 'Ground truth' columns display schematic diagrams of neural network architectures (transformers/attention blocks) rather than the 'multi-dimensional task structure' described in the input.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology from the multi-agent systems and computer vision literature, which creates a barrier for non-specialist readers. The central concept of a "harness" is introduced in the Abstract and Introduction without a clear, plain-English definition, forcing the reader to infer its meaning from context. Similarly, terms like "directive diagnostics," "typed edits," and "convergence judge" are used frequently but lack immediate clarification. While these terms are precise for experts, they obscure the core mechanism for a broader audience. The acronym "VLM" is used without expansion in the

Abstract. Additionally, phrases like “agentic baseline” and “per-dimension scores” could be simplified to “multi-agent baseline” and “scores for each quality aspect” respectively to improve readability. The paper would benefit from defining these terms upon first use or replacing them with more accessible alternatives.
Action items.

- [writing] The manuscript relies heavily on specialized terminology from the multi-agent systems and computer vision literature, which creates a barrier for non-specialist readers. The central concept of a “harness” is introduced in the Abstract and Introduction without a clear, plain-English definition, forcing the reader to infer its meaning from context. Similarly, terms like “directive diagnostics,” “typed edits,” and “convergence judge” are used frequently but lack immediate clarification. While these

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument for the “harness” approach, effectively identifying the limitations of current single-pass or free-text iterative methods (prompt degradation, lack of structured feedback) and proposing a solution (typed edits, directive diagnostics) that directly addresses these premises. The causal chain from the problem definition to the method design is sound.

However, there are specific inconsistencies between the textual claims and the reported data in the results section that undermine the logical consistency of the empirical validation.

1. Data Discrepancy in Main Results (Section 5.1): The text claims that PaperBanana’s performance gain shrinks from 22.60 points on PaperBanana-Bench to 8.10 points on CrafterBench. A review of Table 1 reveals that PaperBanana (w/ Nano Banana 2) scores 33.73 on PaperBanana-Bench and 28.00 on CrafterBench. The difference is 5.73 points, not 8.10. Furthermore, the text implies a comparison of “gain over backbone,” but the numbers cited do not align with the table’s “Overall” scores or the calculated deltas for PaperBanana. This suggests a calculation error or a misstatement of the baseline performance in the text, which breaks the logical link between the evidence (Table 1) and the conclusion (PaperBanana fails to generalize).

2. Incorrect Qualitative Claim in Ablation (Section 5.2): The text states that when the “plan exploration” mechanism is removed, “Readability suffers the most among four quality dimensions.” Table 2 shows that for the “w/o plan exploration” row, Faithfulness drops by 9.76 points (38.18 to 28.42), while Readability drops by 9.47 points (47.77 to 38.30). Faithfulness actually suffers a larger drop than Readability. The claim that Readability suffers the most is factually incorrect based on the provided data, creating

a contradiction between the textual analysis and the numerical evidence.

3. Ambiguity in Dataset Composition (Section 3.3): The text states that sketch and key-element tasks are drawn entirely from academic figures (totaling 70 samples). Table 1 lists 140 academic figures in total. While the math allows for the remaining 70 academic figures to be in other tasks, the text does not explicitly clarify this distribution, leaving a gap in the logical description of how the benchmark is constructed.

These issues do not invalidate the core hypothesis but require correction to ensure the conclusions drawn from the experiments are logically supported by the data presented.
Action items.

- [science] Section 5.1 claims PaperBanana's gain shrinks from 22.60 to 8.10 points. Table 1 shows scores of 33.73 and 28.00, a difference of 5.73. The text's numerical claim contradicts the table data. Verify calculations.
- [writing] Section 5.2 states 'Readability suffers the most' when plan exploration is removed. Table 2 shows Faithfulness drops 9.76 points vs Readability's 9.47. The text contradicts the table data.
- [writing] Section 3.3 claims sketch/key-element tasks are 'entirely from academic figures' (70 total). Table 1 lists 140 academic figures. The text omits how the remaining 70 academic figures are distributed, creating ambiguity in the dataset composition logic.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend beyond the immediate scope of the provided experimental data, specifically regarding the universality of the "harness" abstraction and the novelty of the pipeline.

First, the Abstract and Contributions section claim that Crafter and Editor form the "first end-to-end generation-to-editing pipeline for scientific figures." This is an overreach. The Related Work and Introduction explicitly cite and discuss prior systems like AutoFigure-Edit and Edit-Banana, which also attempt to bridge generation and editing (raster-to-vector). While the paper argues these prior methods are "preliminary" or "limited," claiming to be the "first" ignores the existence of these direct predecessors. The claim should be refined to highlight the *harness-based* nature or the *performance* advantage rather than absolute novelty of the pipeline concept.

Second, the paper repeatedly asserts that the architecture generalizes "across figure types and input conditions without architectural changes" (Abstract, Section 1, Section 3.2). While the experiments show generalization across three specific figure types (academic, poster, infographic) and four input conditions within the scientific domain, the data does not support a claim of generalization to *any* diverse input or figure type. The "Intent Reasoner" and "Plan Generator" agents are implicitly trained or prompted with scientific figure semantics. Without evidence of the system working on non-scientific diagrams (e.g., architectural blueprints, biological cell maps, or general artistic illustrations) without prompt re-engineering, the claim of "without architectural changes" for the broader domain of "diverse inputs" is an extrapolation.

Finally, the Conclusion states, "we expect the same pattern to extend to structured-output domains beyond scientific figures." This is a speculative claim not grounded in the paper's results. The entire evaluation is confined to the scientific figure domain. While a reasonable hypothesis, presenting it as an expected outcome of the current work overstates the evidence. The authors should qualify this as a future direction or remove the definitive expectation.

These issues are primarily matters of precise wording and scope definition rather than fundamental scientific flaws, but they currently inflate the perceived contribution beyond what the data strictly justifies.

Action items.

- [writing] The claim of being the 'first end-to-end generation-to-editing pipeline' (Abstract) overreaches given prior work like AutoFigure-Edit and Edit-Banana. Qualify as 'first harness-based' or 'state-of-the-art'.
- [writing] The assertion that the architecture generalizes 'without architectural changes' (Abstract) is unsupported outside the scientific domain. The agents are implicitly tuned for scientific figures; remove the absolute claim or limit scope to scientific figures.
- [writing] The Conclusion's expectation that the pattern extends to 'domains beyond scientific figures' is speculative and unsupported by data. Rephrase as a future direction or hypothesis.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a technical framework for scientific figure generation but requires minor revisions to fully address safety and ethics standards regarding human subjects and data provenance.

First, the Human Evaluation described in Appendix \ref{app:human-eval} involves human annotators. The text states they were compensated at "$25 per hour" but fails to provide the total duration of the task, the total compensation per participant, or, crucially, the IRB (Institutional Review Board) approval status. For any research involving human subjects (even for annotation), standard ethical practice requires either explicit IRB approval or a formal determination of exemption by the authors' institution. The manuscript must explicitly state the IRB protocol number or the specific exemption category under which this study was conducted to ensure compliance with ethical research standards.

Second, the Data Construction for \CrafterBench (Section \ref{sec:bench-data}) involves scraping figures from arXiv, conference proceedings (CVPR, ICLR), and research blogs. While this is generally considered fair use for benchmark creation, the paper should include a brief statement confirming that the authors verified the absence of Personally Identifiable Information (PII) in the scraped figures (e.g., author photos, contact details) and that the dataset construction adheres to the terms of service of the source platforms. This clarifies the ethical handling of third-party intellectual property and privacy.

Finally, the evaluation pipeline relies heavily on proprietary, closed-source models (Gemini 3.5 Flash, GPT-5) as judges (Section \ref{sec:bench-judge}). While not a fatal flaw, the paper should briefly acknowledge the opacity and potential bias inherent in using black-box models for scientific evaluation. A sentence in the Limitations section (Appendix \ref{app:limitations}) addressing how the authors mitigated or plan to mitigate the risk of proprietary model bias influencing the reported "State-of-the-art" claims would strengthen the ethical rigor of the evaluation.

These are primarily documentation gaps rather than fundamental safety failures, but they are necessary for a complete ethical review.

Action items.

- [writing] The human evaluation section (Appendix Human Evaluation) states annotators were compensated at $25/hour but does not specify the total hours worked, total payment per annotator, or the ethical approval (IRB) status for this human-subject research. Explicitly state the IRB exemption/approval number or confirm the study was deemed exempt by the authors' institution.
- [writing] The benchmark construction (Section 4.1) involves scraping figures from arXiv, conference posters, and blogs. While likely fair use for research, the paper should explicitly address data privacy and copyright considerations, confirming that no personally identifiable information (PII) was extracted from the source figures and that the dataset usage complies with the source platforms' terms of service.
- [writing] The system relies on closed-source, proprietary models (Gemini, GPT) for both generation and evaluation (Section 5, Appendix A). The paper should briefly discuss the potential for bias in these proprietary models affecting the evaluation scores and the lack of transparency in the evaluation pipeline as a safety/ethics limitation.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a novel “harness” architecture for scientific figure generation, supported by a new benchmark (CrafterBench) and extensive experiments. However, the scientific evidence supporting the robustness of the claims requires strengthening in three specific areas regarding statistical rigor and evaluation validity.

First, the validation of the automated VLM judge relies on a human evaluation study described in the Appendix (Section “Human Evaluation”). The study reports a Cohen’s kappa of 0.58 based on only 60 pairwise comparisons. For a benchmark of 279 samples spanning diverse tasks and figure types, a sample size of 60 is statistically underpowered to establish the judge’s reliability as a proxy for human preference. The confidence intervals around this agreement metric are likely wide, and the result may not generalize to the full dataset. The authors should either expand the human study to a statistically robust size (e.g., >200 samples) or provide a formal power analysis justifying why 60 samples are sufficient to validate the metric for the entire benchmark.

Second, the results in Table 1 show that open-source baselines (GLM-Image, Qwen-Image) achieved a 0.00% win-rate across all dimensions on CrafterBench. While these models may be weaker, an absolute zero score across 279 samples and multiple dimensions is highly suspicious and suggests a potential systematic failure in the evaluation protocol (e.g., the VLM judge failing to parse or score the specific output format of these models) rather than a genuine lack of capability. Without qualitative examples or error logs explaining this total failure, the claim that the harness “substantially outperforms” these baselines is difficult to verify. The authors must clarify whether this is a true performance gap or an artifact of the evaluation setup.

Third, the ablation study (Table 2) demonstrates performance drops when removing specific components (ranging from 5.04 to 8.90 points). However, the paper does not report variance (standard deviation) or statistical significance tests (e.g., paired t-tests or bootstrap confidence intervals) for these differences. Given the stochastic nature of LLM-based generation and the relatively small benchmark size, it is unclear if these drops are statistically significant or simply due to random seed variance. The authors should run multiple seeds for the ablation conditions and report statistical significance to confirm that each mechanism’s contribution is robust.

Addressing these points will significantly strengthen the scientific validity of the paper’s central claims regarding the harness’s effectiveness and the reliability of the proposed evaluation metrics.

Action items.

- [science] The human evaluation study (Appendix, Section ‘Human Evaluation’) reports a Cohen’s kappa of 0.58 on only 60 samples. This sample size is statistically underpowered to validate the VLM judge as a reliable proxy for the full 279-sample benchmark. The authors must either expand the human study to a statistically significant size (e.g., >200 samples) or provide a power analysis justifying the current N.

- [science] Table 1 reports a 0.00% win-rate for open-source baselines (GLM-Image, Qwen-Image) on the 279-sample CrafterBench. This absolute failure across all dimensions suggests a potential evaluation protocol mismatch (e.g., the VLM judge failing to parse open-source outputs) rather than a genuine performance gap. The authors must provide qualitative examples or error logs explaining why these models scored zero to rule out a systematic evaluation artifact.
- [science] The ablation study (Table 2) removes mechanisms one at a time but does not report variance (standard deviation) or statistical significance tests (e.g., paired t-tests) for the performance drops. Given the small benchmark size (n=279) and the stochastic nature of LLM-based generation, the authors must demonstrate that the observed drops (e.g., 5.04 to 8.90 points) are statistically significant and not due to random seed variance.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis supporting the claims in this paper is insufficient for a rigorous scientific contribution. While the experimental design (ablations, baselines) is conceptually sound, the reporting of results lacks necessary statistical validation.

First, the primary evaluation metric relies on a VLM-as-judge protocol. The paper reports aggregate win-rates (e.g., Table 1) and a single point estimate for human agreement (72%, $\kappa=0.58$ in Appendix). However, no confidence intervals (CIs) or standard errors are provided for these metrics. Given the inherent stochasticity of VLM judges and the finite sample sizes ($n=279$ for CrafterBench, $n=292$ for PaperBananaBench), it is impossible to determine if the reported margins (e.g., the −16.61 point lead over the strongest baseline) are statistically significant or within the noise floor of the evaluation method. The authors should report 95% CIs for all win-rates and perform paired significance tests (e.g., Wilcoxon signed-rank test) between the proposed method and baselines.

Second, the ablation study (Table 2) claims that "every mechanism contributes independently" based on point drops ranging from 5.04 to 8.90. Without statistical testing, these differences could be artifacts of random variation, especially given the non-deterministic nature of the generation and evaluation pipeline. The authors must demonstrate that the performance drop upon removing a component is statistically significant (e.g., $p < 0.05$) to support this claim.

Third, the human evaluation in the Appendix (Section A.8) uses a small sample size ($N=60$) to validate the automated judge. The reported Cohen's κ of 0.58 is a point estimate; a confidence interval for κ is essential to understand the precision of this agreement metric. Furthermore, the claim that the metric is a "reliable proxy" is weak without a power analysis or a demonstration that the sample size was sufficient to detect a meaningful correlation.

Finally, the paper mentions “random seeds” for the harness loop but does not report the variance across multiple runs. For a system involving iterative refinement and multiple agents, reporting the mean and standard deviation over at least 3-5 independent runs per sample (or per task subset) would provide a much stronger basis for the reported performance differences. The current presentation of single-run results is insufficient for reproducibility and statistical confidence.
Action items.

- [science] The VLM-as-judge evaluation lacks statistical rigor. The paper reports mean win-rates and Cohen’s kappa (0.58) but omits confidence intervals, standard errors, or significance tests (e.g., paired t-tests or Wilcoxon signed-rank) to validate that the observed margins (e.g., −16.61 points) are not due to random variance in the judge’s scoring.
- [science] The ablation study (Table 2) reports point drops (e.g., -8.90) without statistical validation. Given the small sample size (n=292 for PaperBananaBench), the authors must report whether these differences are statistically significant to support the claim that ‘every mechanism contributes independently’.
- [science] The human evaluation (Appendix) uses a small sample (N=60) to validate the VLM judge. The reported agreement (72%) and kappa (0.58) lack confidence intervals. A binomial proportion confidence interval is required to assess the reliability of this proxy metric.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling technical contribution, but the writing quality requires minor revisions to ensure professional polish and grammatical precision. While the overall flow is logical and the structure is clear, several specific errors detract from the reading experience.

First, there are isolated typos and grammatical slips. In the Introduction (Section 1, paragraph 2), the word “viusal” appears instead of “visual” in the phrase “reference viusal elements.” In Section 2 (Related Work), the sentence “Evaluation method for scientific figure generation remains as narrow as systems it measures” contains a subject-verb agreement error and missing articles; it should read “Evaluation methods… remain as narrow as the systems they measure.” Similarly, in Section 5.2 (Ablation and Analysis), the sentence “Together, both ablations confirms that…” incorrectly uses the singular verb "confirms" for the plural subject "ablations."

Second, there is an inconsistency in the naming of the benchmark. The paper consistently refers to the benchmark as "CrafterBench" (e.g., in the Abstract, Section 3, and Section 5), but Table 1's caption and the text in Section 5.1 occasionally use "CraftBench." This inconsistency should be resolved to maintain a unified terminology throughout the document.

Finally, while the LaTeX source contains some commented-out code and template instructions (e.g., the NeurIPS track selection comments), these do not affect the final compiled text but should be cleaned up in the final version to avoid confusion. The prose itself is generally strong, but addressing these specific points will elevate the manuscript to the expected standard for a top-tier conference.

Action items.

- [writing] Correct the typo 'viusal' to 'visual' in the Introduction, second paragraph, where the text discusses researchers iterating from rough sketches.
- [writing] Fix the grammatical error 'Evaluation method for scientific figure generation remains as narrow as systems it measures' in Section 2. Change to 'Evaluation methods... remain as narrow as the systems they measure'.
- [writing] Resolve the subject-verb agreement error in Section 5.2: 'Together, both ablations confirms that...' should be 'confirm that...'.
- [writing] Standardize the capitalization of 'CraftBench' in Table 1 caption and Section 5.1 to match the consistent usage of 'CrafterBench' elsewhere in the paper.